# HeatCast NCR: Speaker Script

### Final presentation, about 9 to 10 minutes plus Q&A

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## Who says what

| Slide | Title | Presenter | Time |
|---|---|---|---|
| 1 | Title | Edrian | 20 seconds |
| 2 | Our question | Edrian | 30 seconds |
| 3 | The answer: 82% | Edrian | 20 seconds |
| 4 | The problem and the data | Harry | 50 seconds |
| 5 | What the data showed | Marc | 1 minute |
| 6 | A fair test, like an exam | Harry | 1 minute |
| 7 | Which mistake is worse? | Marc | 40 seconds |
| 8 | Three models vs. copy-today | Harry, Edrian, Marc | 1 minute 15 (about 25 seconds each) |
| 9 | Danger days caught | Marc | 25 seconds |
| 10 | Final exam result | Marc | 50 seconds |
| 11 | Live demo | Edrian | 1 minute 45 |
| 12 | Conclusion | Harry | 50 seconds |
| 13 | Thank you / questions | All | Q&A |
| 14 | Backup: key terms | Anyone, if asked | only during Q&A |

Approximate totals: Edrian 3:20, Harry 3:05, Marc 3:20 (each includes about 25 seconds on slide 8 for your own model).

## Before the talk

- Open heatcast-ncr.streamlit.app at least 5 minutes early; the first load can take about a minute. Keep it in a second tab.
- Have the backup ready: the screenshot on slide 11, or run the app locally with `streamlit run app/streamlit_app.py`.
- Slide 14 is a backup for questions only; stop at slide 13.
- Rehearse once with a timer. If you run long, shorten slide 5 (say only the first two findings) and the demo (skip step 3).

## The script, slide by slide

### Slide 1. Title

**Edrian** · 20 seconds

Good morning. We are Edrian, Harry and Marc, and this is HeatCast NCR. Our question: can machine learning predict tomorrow’s heat level in Metro Manila better than just copying today’s level? Let’s start with a simple example.

### Slide 2. Our question

**Edrian** · 30 seconds

Here is our question as an example. Today Manila's heat index reached 44 degrees, which is Danger. Tomorrow can go three ways: hotter, Extreme Danger; the same level, Danger again; or cooler. HeatCast NCR has to pick one of these, for each of the 16 cities, every day. So which one happens most often?

### Slide 3. The answer: 82%

**Edrian** · 20 seconds

Most often, the same level. Each square here is one percent of the city-days in 2025. In 82 out of every 100, tomorrow's level was the same as today's. So the simplest forecast, copy today, is already right most of the time. We call it the copy-today rule, and the real question of our project is whether machine learning can beat it.

### Slide 4. The problem and the data

**Harry** · 50 seconds

Thanks, Edrian. The heat index is how hot it feels when you combine temperature and humidity. PAGASA groups it into five levels; Danger starts at 42 degrees. Schools, employers of outdoor workers and city offices need to know a day ahead whether tomorrow will reach Danger. Our data covers the 16 cities of Metro Manila from 2021 to 2025: 29,168 rows, one per city per day. Each row has five weather readings about today, and the answer is tomorrow's level in the same city. The weather comes from Open-Meteo's hourly estimates.

### Slide 5. What the data showed

**Marc** · 1 minute

Before building models, we explored the data, using only the training years. Three findings changed our decisions. One: Danger is rare, 8% of city-days, while 78.5% are Extreme Caution. So a lazy model that always says Extreme Caution is already 78% correct. That is why we do not judge the models by percent correct. Two, in the chart: the red part is Danger. It peaks from April to June, nearly one in three May days, and it never happens from December to February. So we split the data by time, the way a real forecast works. Three: cities are different. Malabon, Navotas and Valenzuela have Danger on 17% of days, Muntinlupa and San Juan on only 2%. So each city keeps its own rows.

### Slide 6. A fair test, like an exam

**Harry** · 1 minute

To compare the models fairly, think of an exam. All three models studied the same days, 2021 to 2024, and took the same final exam, 2025, which we opened only once at the very end. While choosing settings, each model took five practice quizzes: it studies earlier days and is quizzed on the days right after, never the other way around, because we cannot use the future to predict the past. The score counts Caution, Extreme Caution and Danger equally, from 0 to 1. Our rule, fixed before the final exam: first, a model must catch more Danger days than the copy-today rule; then the best score wins. Technically this is a chronological split with time-series cross-validation, and the score is macro F1.

### Slide 7. Which mistake is worse?

**Marc** · 40 seconds

Every forecast is sometimes wrong, and it can be wrong in two ways. One: it misses a real Danger day, so nobody prepares, students play outside at 44 degrees, and someone may get sick. Two: a false alarm, so activities are moved indoors on a day that was not that hot. We decided the first mistake is worse: a missed Danger day can hurt people, while a false alarm only costs a change of plans. That is why our rule asks a model to catch more Danger days first. Keep this in mind for the results.

## Slide 8. Three models vs. copy-today

**Harry, Edrian, Marc** · 1 minute 15 (about 25 seconds each)

HARRY: Here are the practice-quiz results. Top chart: the score. Bottom chart: the share of Danger days each one caught. Look at the copy-today rule at the top: its score, 0.679, is the highest. None of our models beat it on score. My model, Logistic Regression, draws straight dividing lines between levels. It was careful, but it caught only 28% of Danger days, even fewer than copying today.

EDRIAN: Random Forest is 500 decision trees that vote. We told it that Danger days matter more, and it caught 67% of them, more than copy-today, with the best score of our three. By our rule, it was selected.

MARC: Gradient Boosting builds trees one after another, each fixing the last one's mistakes. It was almost tied with Random Forest, 0.643 against 0.653, so either would have worked.

## Slide 9. Danger days caught

**Marc** · 25 seconds

Now the final exam, 2025, which we opened only once. There were 347 Danger days across the 16 cities. Copying today warned about 207 of them, 60%. Our Random Forest warned about 293, 84%. That is 86 more Danger days with a warning the day before.

## Slide 10. Final exam result

**Marc** · 50 seconds

Here is the full picture. Catching 293 Danger days has a price: when it warned Danger, it was right only 36% of the time, about two false alarms for every real Danger day. And its overall score, 0.61, is below copy-today's 0.68. Remember our choice: we said missing a Danger day is worse than a false alarm. So it is not a better all-round forecaster, but it is a good early warning. Edrian will show it live.

## Slide 11. Live demo

**Edrian** · 1 minute 45

BEFORE THE TALK: open heatcast-ncr.streamlit.app; the first load can take about a minute. If the internet fails, show this screenshot, or run it locally with: streamlit run app/streamlit_app.py.

Let's try it with Manila. (Choose it at the top; the city lights up blue on the map.) Say today was very hot there: heat index 45.5, lowest temperature 29, rain 0, pressure 1008, sunshine 7000. Click the button. It predicts the level for tomorrow with the probability of each level, and the city takes that level's colour on the map. It also reminds users it is a decision aid, not a PAGASA advisory. Now a mistake on purpose: rain of minus 5. The app refuses it and shows the accepted range. And if we clear the form and click, it names every missing value instead of guessing. This is the same saved model that took the final exam, and the input checks are covered by 51 automated tests.

## Slide 12. Conclusion

**Harry** · 50 seconds

So, our answer: no, none of our models beat simply copying today's level on overall score. Tomorrow's heat is mostly today's heat. But machine learning adds something specific: when we told the Random Forest that Danger days matter more, it caught 84% of them in 2025, so we deployed it as an early warning. Its limits: the weather values are estimates, not station readings, and we used only five years. Next we would add the month and the previous days, a weather forecast for tomorrow, and more years. Thank you.

### Slide 13. Thank you / questions

**All** · Q&A

Thank you. We are happy to take questions. The next slide is a backup with plain meanings of the technical words, in case a question uses them.

### Slide 14. Backup: key terms

**Anyone, if asked** · only during Q&A

Use this slide only if a question needs it. Likely questions: Why not percent correct? Because Danger is rare; always saying Extreme Caution is already 78% correct. Why split by time? Because we forecast the future; a random split would let a model peek at the same days. Why Random Forest over Gradient Boosting? Our rule picked the best score among the models that catch more Danger days; the two were almost tied. Why did no model beat copy-today? Tomorrow's level is the same as today's on most days, and the models only see today's values.

## Likely questions and short answers

**Why didn't you use accuracy (percent correct)?**
Danger is rare. A model that always says Extreme Caution is already about 78% correct but never warns of Danger. So we used a score that counts each level equally (macro F1) and the share of Danger days caught (Danger recall).

**Why split by time instead of randomly?**
We forecast the future. A random split would put days from the same week in both training and testing, so the model would get an unfair peek. Training on 2021 to 2024 and testing on 2025 is how it would really be used.

**Why did no model beat copying today's level?**
Tomorrow's level is the same as today's on about 82% of days, and our models only see today's five values. Adding the month, the previous days and a weather forecast are our main next steps.

**Why Random Forest and not Gradient Boosting?**
Our rule, fixed before the final exam: first catch more Danger days than copy-today, then the best score. Random Forest scored 0.653 and Gradient Boosting 0.643; that gap is smaller than the variation between quizzes, so either would have worked.

**What does 'told Danger matters more' mean?**
Balanced class weights: during learning, each rare Danger day counts more, so the model pays attention to it instead of always predicting Extreme Caution.

**Isn't 36% precision bad?**
It means about two false alarms for every real Danger day. For schools and outdoor work, a false alarm costs a change of plans while a missed Danger day can hurt people, so we accept the trade-off and say so clearly.

**Where does the data come from? Is it PAGASA data?**
Hourly ERA5 weather estimates from Open-Meteo (about 25 km grid), turned into a heat index with the US National Weather Service formula. They are not PAGASA station readings, which is one of our limits.

**Why only 2021 to 2025?**
We kept the last five years to keep the dataset simple and recent. The original file goes back to 2015; using it all is a next step.

**Can it predict Extreme Danger?**
No. Extreme Danger never happened in 2021 to 2025, so the model has never seen it.